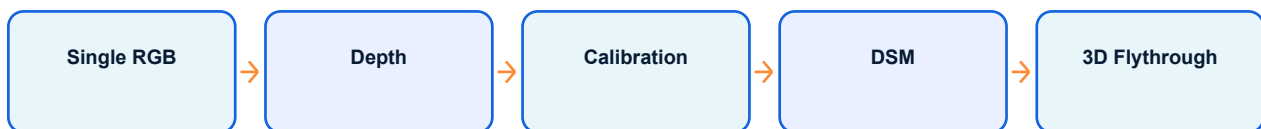


TerraFly SIH Winning and Presentation Playbook

Winner evidence, proof package, six-person script, research ownership, Q&A; and demo choreography



PURPOSE

This is not a promise of winning. It converts verified winner patterns and the exact 50/50 PS rubric into work the team can demonstrate, measure and defend.

Team: TerraFly

Problem Statement: SIH26175 - DepthWizard - Single-View Height Estimation and 3D Flythrough

A practical guide for an honest, measurable, judge-ready system.

1. The honest verdict

THE MAIN DECISION

Build the smallest complete end-to-end system that tells the truth about its output. A polished relative-height map is better than a fake metric DSM. A measured metric DSM is better than ten unverified AI claims.

The public record does not provide one central archive containing every 2025 winning deck, judge score sheet and explanation of why each team won. Therefore, this guide does not invent a secret formula. It uses a small set of verifiable winner accounts, public proof packages, official event reports and your exact SIH26175 text. Where a conclusion is an inference, it is labelled as a pattern rather than a fact.

Your college tier is not a technical variable in SIH26175. The controllable variables are evidence, clarity, reliability, teamwork, response to feedback and how well the prototype matches the stated evaluation criteria.

Winning is not controllable. Becoming the team with the clearest evidence is controllable.

Evidence labels

Label	Meaning	How to use it
Official	Organizer, ministry, official documentation or the supplied PS file.	Use for rules, requirements, policy and technical definitions.
Winner account	A named winning team or a public winning repository.	Use for process patterns, not universal judging rules.
Finalist / internal	Useful experience but not proof of national-winning causation.	Use as a practical signal and label it honestly.
Inference	A conclusion drawn across several sources.	Present it as our strategy, not as an official SIH rule.

LOCAL SCORE-SHEET SIGNAL

The supplied SIGCE internal result lists the highest scores as 72, 72 and 70. Their problem statements were specific, operational and easy to visualize: real-time scam detection, offline emergency evacuation and edge-based road-sign perception. The sheet does not show prototypes or judge reasoning, so we cannot claim those themes caused the scores. The safe lesson is to make TerraFly equally concrete: input, processing, measurable output, user and operating context.

2. What verified 2025 winners did differently

These cases are different domains, but the repeated behaviours are useful for TerraFly.

Case A - Team TwinX, national winner for a MathWorks problem statement

Observed behaviour	Why it mattered	TerraFly translation
Chose a visible real problem with room for innovation, not simply the easiest familiar topic.	The team could explain user pain and practical value before discussing technology.	Open with the cost and sensor-dependency of stereo, LiDAR and InSAR, then show the single-image alternative and its limits.
Applied design thinking and split a broad PS into components.	A complex challenge became a sequence of testable subsystems.	Separate input validation, depth inference, calibration, geospatial export, mesh generation, viewer and evaluation.
Built an end-to-end workflow with Indian-domain behaviour and export into established tools.	The outcome was usable beyond one isolated AI model.	Show upload-to-output flow, Indian landscape cases, GeoTIFF export and a standalone 3D viewer.
Used expert feedback, refined usability and changed the system after judge feedback.	The product visibly improved between evaluations.	Maintain a judge-feedback log and demonstrate one change at the next round.
Worked according to team strengths and kept building through the finale.	Ownership and coordination were part of execution.	Give every member one real artifact and one technical question they can answer.

Source: MathWorks Student Lounge winner account [W1].

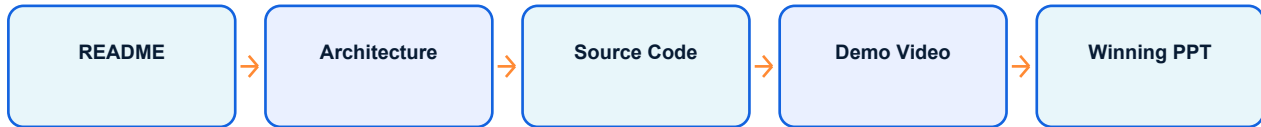
Case B - voice-based bus tracking and ticketing national winner

Observed behaviour	TerraFly lesson
Started from a personal, easy-to-understand mobility pain point.	Translate remote-sensing language into one user story before saying DSM or rDSM.
Built multilingual voice-enabled real-time booking and tracking for smaller cities.	Domain fit matters: show why the system is useful under Indian coverage, cost and connectivity constraints.
The team understood the problem before coding and coordinated roles during 36 hours.	The six-person script should sound like one system, not six separate assignments.
The published role split included ideation, research/presentation, frontend, integration and calm execution.	Do not give fake roles. Each member should own a verifiable deliverable.

Source: India Today winner profile [W2].

3. The strongest public proof-package example

Team Ourobonics / Kritrim published a public SIH 2025 winner repository containing a README, code, requirements, architecture diagram, team information, demo video and winning PPT. That does not prove every winner used the same package, but it demonstrates a powerful principle: judges can inspect more than promises.



What TerraFly should hand over

Artifact	Minimum acceptable version	Owner
Live application	Upload image, run one complete path, label output truthfully, open 3D viewer.	Core development + integration
90-second demo video	Screen recording with voice-over; shows input, processing, output, metrics and export. Keep it offline on two laptops and a phone.	Product / QA
README	Problem, setup, one-command launch, sample inputs, outputs, limitations, licenses.	Documentation owner
Architecture diagram	One page: frontend, API, model, calibration, raster processing, mesh/viewer and output files.	Om + core developer
Validation report	Dataset split, units, alignment method, RMSE, MAE, correlation, timing, landscape results and failure cases.	Data / validation
Source and requirements	Pinned versions, clean folders, no secret keys, repeatable environment.	Om
Presentation and one-page handout	Same claims, same numbers, same terminology; no unmeasured accuracy claim.	Saman + Parth
License / provenance ledger	Every dataset, model and asset with source, license and permitted use.	Research + validation

Source: public winner repository [W3].

THE DIFFERENCE

A deck says what you intend to build. A proof package lets a judge verify what exists, how it was tested and whether the team owns the work.

4. Repeatable winner pattern - and what is only advice

Pattern	Evidence strength	TerraFly action
Understand and narrow the problem before choosing technology.	Repeated in winner accounts and reflected in the PS itself.	Define two exact modes: PNG/JPG produces rDSM; GeoTIFF plus reference produces metric DSM.
Show an end-to-end usable workflow.	Strong winner pattern.	Run input to 3D and export without switching to disconnected notebooks.
Use feedback to improve between judging rounds.	Explicit in the TwinX account.	Write every judge comment; assign owner and show the change.
Use diagrams, screenshots, MVP link and short demo.	Winner-authored advice; not official scoring language.	Use it because it improves comprehension, not because it guarantees selection.
Every member speaks and owns a subsystem.	Finalist and winner-role evidence; not a universal published rule.	Use six short connected segments and a truthful ownership ledger.
Keep one dramatic AI claim.	No reliable evidence; risky advice.	Reject it. Use measured claims and label uncertainty.

Winner-authored shortlist advice includes using diagrams instead of essays, a working MVP, a short demo video and a simple 15-second explanation test [W4]. A 2025 finalist account similarly emphasizes proof, architecture, clean flow and every member speaking [W5]. Treat both as experience, not organizer policy.

The TerraFly one-sentence explanation

SAY THIS

TerraFly turns one authorized RGB satellite or aerial image into an elevation surface and navigable 3D scene. For ordinary PNG/JPG files it produces relative shape only; for a georeferenced GeoTIFF it uses SRTM or known control heights to calibrate that shape into metric elevation, then reports measured error against reference data.

If a non-technical person cannot repeat that explanation after hearing it once, simplify again.

5. Match the exact SIH26175 rubric

The supplied official spreadsheet assigns 50 percent to DSM estimation accuracy and validation, and 50 percent to rendering quality and user experience. A beautiful viewer with no measured DSM is half a solution. A good model hidden in a notebook is also half a solution.

Rubric half	What the judges can inspect	Minimum judge-ready evidence
DSM accuracy and validation - 50%	RMSE, MAE, correlation, stability across urban, sparse, hilly and forested scenes.	Held-out reference raster; aligned CRS/resolution; metrics in metres for metric mode; per-landscape table; baseline; failure case; residual map.
Rendering and UX - 50%	Projection accuracy, visual fidelity, navigation, intuitiveness, stability and standalone deployment.	Correct texture alignment; first-person and orbit controls; height/slope inspection; progress/error states; export; offline launch; recorded fallback demo.

INPUT ROUTING



NON-NEGOTIABLE TRUTH LABEL

Relative output has no metres. Metric height is shown only after geospatial alignment and calibration. Keep a permanent mode badge in the UI: RELATIVE / NO METRIC CLAIM or CALIBRATED METRIC / REFERENCE: [name].

Source: supplied SIH 2026 Official PS.xlsx, row SIH26175 [P1].

6. Claim ladder - what you may say

Claim level	Required proof	Safe example
Implemented	A working route or screen.	The app accepts PNG, JPG and GeoTIFF and generates a 3D mesh.
Measured	A recorded test on identified data.	On [dataset/split], the calibrated DSM achieved RMSE [X] m, MAE [Y] m and correlation [Z].
Compared	Same split and alignment for a baseline.	Calibration reduced MAE from [A] m to [B] m compared with the uncalibrated baseline.
Generalizes	Separate landscapes and held-out locations.	Performance was tested separately on urban, sparse, hilly and forested subsets.
Operational	Install/run evidence, timing and failure recovery.	The packaged app launched offline on two Windows machines and processed a [size] tile in [time].
Future / potential	No claim of current adoption.	The export format is designed for potential integration into authorized ISRO/NRSC-style workflows.

Never say

- "This produces exact building heights from any image."
- "The model is 95 percent accurate." Accuracy is undefined here unless you state dataset, metric, units and split.
- "ISRO will use this." You have no authority to claim adoption.
- "India bans all foreign mapping, so Google imagery is bad." The more accurate position is that high-fidelity 3D coverage varies and depends on aerial acquisition, licensing and local regulations [I1-I4].
- "It is secure because it is offline." Offline reduces network exposure, but file validation, local access and dependency risk still matter.

USEFUL POSITIONING

TerraFly is not a replacement for survey-grade LiDAR or stereo photogrammetry. It is a fast first-look and decision-support pipeline when only one authorized optical view is available, with explicit calibration, validation and uncertainty.

7. Six-person presentation - recommended flow

This script assumes approximately 7 minutes plus Q&A.; Shorten it proportionally if your college gives less time. Never read the slide word-for-word.

Time	Speaker	Job in the story	Visual
0:00-0:45	1. Team lead / core developer	Hook, user pain and one-sentence solution.	Before/after hero: RGB to DSM to 3D.
0:45-1:35	2. Saman - research and presentation	Who needs it, why existing acquisition is costly, two input modes.	Problem, stakeholders and mode split.
1:35-2:25	3. Parth - product and Q&A;	Gap, novelty and practical value.	Comparison / gap matrix.
2:25-3:25	4. Om - AI and reliability	Inference pipeline, CUDA, preprocessing, relative-depth honesty.	Technical architecture.
3:25-4:25	5. Data and validation lead	Calibration, geospatial alignment, metrics and uncertainty.	Calibration plot + metrics table.
4:25-5:25	6. Product, viewer and QA lead	3D mesh, flythrough, exports, offline behaviour and testing.	Viewer screen + test evidence.
5:25-6:40	Live demo - speakers hand off	Run the two-mode story with a prepared sample.	Application, not slides.
6:40-7:00	Team lead	Measured close and specific ask.	Three proof points.

8. Exact speaking script - speakers 1 to 3

Speaker 1 - team lead / core developer - 45 seconds

"Good [morning/afternoon]. Today, one accurate 3D terrain model often needs stereo imagery, LiDAR or other specialized sensing. But many real workflows begin with only one optical RGB image. TerraFly is our end-to-end attempt to turn that single authorized image into an elevation surface and a navigable 3D scene. We make one distinction very clear: a normal PNG or JPG produces relative shape, not fake metres; a georeferenced GeoTIFF can be calibrated with SRTM or known control heights to produce metric elevation that we then validate. I will first show why that distinction matters."

Speaker 2 - Saman, research and presentation - 50 seconds

"Accurate elevation supports urban planning, disaster assessment, terrain analysis and reconnaissance. Traditional acquisition can be expensive, sensor-dependent or slow, while single-view depth models are usually trained on ground-level natural images and do not automatically understand overhead scenes or absolute scale. Our users are analysts who need a rapid first-look surface from imagery they are authorized to use. TerraFly therefore has two honest routes. The relative route preserves terrain and structure shape for visualization. The calibrated route reads geographic metadata, aligns an external elevation reference and converts the relative prediction into metres. We do not claim to replace survey-grade sensing; we reduce the gap when only one view is available."

Speaker 3 - Parth, product and Q&A; - 50 seconds

"A pretrained depth model alone is not the solution. Existing models can create visually convincing depth but usually omit geospatial scale, raster export, landscape-wise validation and an operational viewer. TerraFly connects these missing pieces: file validation, monocular depth inference, geospatial calibration, DSM export, a textured mesh, first-person navigation and height or slope analysis. Our novelty is the truthful two-mode workflow and the evidence around it. Every output carries its mode, reference source, units and processing record. The system is designed to run locally so authorized imagery does not need to be uploaded to an external cloud, while the export remains compatible with standard geospatial workflows."

9. Script - AI, validation, viewer and close

Speaker 4 - Om, AI and reliability - 60 seconds

"The pipeline first validates the file, image bands, dimensions and geospatial metadata. We then normalize the RGB image and run a pretrained monocular-depth backbone using PyTorch. On our RTX GPU, CUDA means PyTorch can execute tensor operations on the NVIDIA graphics processor; it does not make the model more accurate, but it can make inference faster. Because the raw model output is scale-ambiguous, we store it as floating-point relative depth and never label it as metres. Post-processing handles invalid pixels, optional tiling and smoothing without hiding the original prediction. We also provide a smaller-model or CPU fallback so a failed GPU setup does not break the complete demo."

Speaker 5 - data and validation lead - 60 seconds

"For a GeoTIFF, we read its coordinate reference system, pixel transform and NoData mask. We reproject and resample SRTM, GCP-derived or reference elevation data onto the same grid. A robust calibration fits a mapping from relative prediction to elevation; all units and residuals are recorded. We then compare the result with held-out reference data using RMSE, MAE and Pearson correlation. The final slide will contain real values only: RMSE [MEASURED VALUE] metres, MAE [MEASURED VALUE] metres and correlation [MEASURED VALUE] on [DATASET AND SPLIT]. We also report urban, sparse, hilly and forested cases separately, because one average can hide failure."

Speaker 6 - product, viewer and QA lead - 60 seconds

"The elevation grid becomes a triangle mesh: each sampled pixel is a vertex, neighbouring vertices form triangles and the original RGB image is mapped over the surface as a texture. The viewer supports orbit inspection, first-person movement and point-based height or slope reading. Relative mode keeps values labelled relative; metric tools unlock only after calibration. We tested upload errors, unsupported bands, missing CRS, unavailable GPU, large images and interrupted jobs. The demo can run offline, exports the DSM and 3D asset, and includes a recorded fallback in case the venue machine or projector fails. Now we will show one prepared path from input to output."

Team lead - final 20 seconds

"TerraFly's value is not an impressive-looking 3D scene alone. It is the complete chain from one image to a labelled elevation product, a navigable scene and evidence of error. Today we can demonstrate [IMPLEMENTED FEATURES], measured on [DATASET], with [PROCESSING TIME] on our test machine. Our next priority is [ONE SPECIFIC LIMITATION]. We welcome one test image or validation condition from the judges."

PLACEHOLDERS ARE DELIBERATE

Do not replace bracketed values until the test has been run and documented. A judge will respect an honest missing result more than an invented number.

10. Demo choreography - make the answer visible

1. Start with a prepared PNG/JPG. Show the input preview, press Generate and point to the permanent badge: RELATIVE / NO METRIC CLAIM.
2. Open the rDSM, rotate the textured mesh and use one relative-height inspection. Say: this is shape, not metres.
3. Load a prepared GeoTIFF. Show detected CRS, pixel size and bounds. If those are missing, the app must stop and explain why.
4. Select the prepared reference source. Show alignment status, calibration method and residual summary.
5. Generate the metric DSM. Click one point or area to display elevation or height in metres together with reference and confidence/residual information.
6. Switch to first-person flythrough. Demonstrate stable navigation, correct texture placement and a slope readout only in metric mode.
7. Open the evaluation panel: RMSE, MAE, correlation, landscape split, runtime, model and test-data version.
8. Export GeoTIFF and GLB/mesh. End by showing the job manifest/README, not by endlessly flying around.

Fallback stack

Failure	Immediate response
Internet unavailable	App and model weights are local; do not depend on a cloud API.
GPU fails	Select small model or CPU mode; have already measured expected runtime.
Live generation too slow	Use a small prepared tile and simultaneously play the 90-second recording.
Projector/viewer issue	Carry screenshots, exported GLB/GeoTIFF, recording and one-page handout on two devices.
Judge provides unknown image	Validate it, state which mode is possible and never force a metric result without reference.

11. Real ownership - six members, six artifacts

Member	Primary ownership	Artifacts to show	Must answer
You - core developer / lead	Architecture, integration, critical path and benchmark decisions.	Integrated app, architecture decision log, runtime benchmark.	Why this design; what is actually implemented; biggest limitation.
Saman - research / presentation	Stakeholders, source ledger, problem framing, policy wording and slide consistency.	Research brief, citation sheet, final narrative and handout.	Who needs it; why now; what evidence supports the problem.
Parth - product / Q&A;	Prior-art comparison, novelty, use cases, judge-question bank and rapid clarification.	Comparison matrix, Q&A; sheet, feedback log.	Why not existing tool; what is different; adoption path.
Om - AI / reliability	Model integration, error handling, CUDA/CPU fallback, tests and reproducible environment.	Model wrapper, test results, requirements lock.	What CUDA does; failure recovery; model limits.
Member 5 - data / validation	Datasets, licensing, alignment, calibration, metrics and failure analysis.	Data card, calibration notebook/script, metric report.	Units; split; RMSE/MAE/correlation; leakage and alignment.
Member 6 - viewer / QA / docs	3D interaction, usability, packaging, demo recording and test checklist.	Viewer, UX test log, packaged build, demo video.	Mesh/texture; standalone mode; tests and exports.

DO NOT INVENT CONTRIBUTIONS

If the last two members join later, give them ownership now and let them produce visible work. Testing is a real role only when it includes test cases, results, bugs found, fixes verified and an artifact the judge can inspect.

12. Research matrix - what the team must know

Research question	Required evidence	Owner
Who is the first user and decision supported?	One use-case flow and one stakeholder interview or authoritative source.	Saman
What is already available and where is the gap?	MiDaS, Depth Anything V2, Depth Pro, GIS/3D options; capability comparison with citations.	Parth
What data may we legally use?	Dataset/model license, attribution, redistribution limits and local copy record.	Member 5 + Saman
What does relative depth mean in this overhead scene?	Raw float output, inversion/sign check, examples and failure modes.	Om
How are CRS, pixel grid and reference raster aligned?	Reprojection/resampling method, NoData mask and visual alignment check.	Member 5
How is scale fitted?	Equation, train/calibration points, robust fitting, residual distribution and held-out evaluation.	Member 5 + you
Does performance change by landscape?	Urban/sparse/hilly/forested subsets and separate metrics.	Member 5
Is the viewer geometrically correct?	Texture/vertex alignment, vertical scale and known-point inspection.	Member 6
Can it run on a normal target machine?	GPU/CPU timing, VRAM/RAM, tile size, crash recovery and packaged launch.	You + Om
What is the security statement?	Local/offline data flow, accepted file types, size limits, path handling and deletion policy.	Om + Parth
What are the known failure conditions?	Clouds, water, shadows, extreme relief, dense high-rise scenes, domain shift and missing metadata.	Entire team

13. Judge questions - scientific truth

Can one image really give true height?

Not by itself in the general case. The PNG/JPG route gives relative shape. Metric elevation requires georeferencing and an external scale/elevation anchor such as aligned SRTM, GCPs or reference data, followed by validation.

Is this just hallucinated depth?

It is a model prediction and may be wrong under domain shift. We preserve raw output, label the mode, quantify residuals, show failure cases and do not present it as ground truth.

Why can SRTM at about 30 m help a higher-resolution image?

It can anchor broad terrain scale and low-frequency elevation, not recover rooftop detail by itself. Fine structure comes from the relative prediction; alignment and held-out residuals reveal whether the fusion helps.

Why RMSE, MAE and correlation together?

MAE is the average absolute error, RMSE penalizes large mistakes more strongly and correlation checks whether predicted shape follows the reference. None alone tells the full story.

What is your accuracy?

Answer only with measured values, units, dataset, split, alignment and date. If not measured, say evaluation is pending and show the protocol rather than inventing a percentage.

How do you avoid data leakage?

Split by location or scene before tuning calibration, keep test locations unseen and never choose calibration parameters using the held-out reference.

What if depth direction is inverted?

Run a documented orientation check against reference points or known high/low areas before scale fitting; record the chosen transform.

What if the GeoTIFF CRS is missing or wrong?

Block metric mode and ask for valid metadata or GCPs. A fabricated CRS would make every height and slope claim unreliable.

14. Judge questions - engineering and differentiation

Why not simply use Depth Anything or MiDaS?

Those are possible backbones. TerraFly is the operational geospatial pipeline around the backbone: input routing, scale calibration, raster metadata, validation, mesh/viewer, exports, provenance and offline packaging.

Why not Google Earth?

Google Earth is a different licensed visualization product and its imagery/3D detail varies by location. TerraFly processes authorized user-supplied imagery locally and produces an auditable derived raster and mesh. It is not a scraper or copy of Google Earth.

Is this better than LiDAR or stereo?

Not as a general accuracy replacement. It reduces input burden and can provide a rapid first-look estimate when those sensors or pairs are unavailable. The comparison should be speed, cost/input availability and measured error, not an absolute superiority claim.

Can it work offline?

The demo build keeps code, weights and assets local. State exactly which setup step may need internet and provide pre-downloaded dependencies. Offline operation reduces cloud transfer but does not replace input security checks.

Will 8 GB VRAM be enough?

For selected small/base pretrained models and prepared tiles, likely yes after benchmarking. Use mixed precision and tiling only after verifying numerical stability; include CPU fallback and measured maximum tile size.

How is slope computed?

From spatial gradients of a metric elevation grid using real pixel spacing after CRS alignment. Relative mode can show relative steepness but must not label degrees or metre-based slope without scale.

What is standalone deployment?

A documented local launcher or packaged application that starts the API and viewer without a developer manually opening notebooks. Demonstrate it on a clean second Windows user profile or machine.

What fails?

Clouds, water, deep shadow, unfamiliar sensors, strong seasonal differences, extreme relief, dense vertical structures and bad metadata. Show at least one failure and the UI warning.

Did one person or AI build everything?

Answer with the truthful artifact ownership table, commit/test history and what each member can explain. AI assistance does not replace validation or responsibility.

15. Presentation rehearsal protocol

Rehearse in four rounds

1. Clarity round: show only the first slide to someone outside computer engineering. Ask them to repeat the problem and the two modes in 15 seconds.
2. Timing round: each speaker delivers without reading. Record the total and cut repetition before cutting proof.
3. Interruption round: Parth or a senior asks a question every 20 to 30 seconds. The current speaker answers in one sentence, then returns to the story.
4. Failure round: disconnect internet, disable GPU, provide a bad file and stop the live demo midway. Practice the fallback without panic.

Three-level answer pattern

Level	Example for calibration
10 seconds	Relative depth has no metre scale, so we align an external elevation reference and fit a scale/offset.
30 seconds	We reproject SRTM/GCP reference to the image grid, fit a robust mapping, keep residuals and evaluate on held-out pixels/areas.
Technical follow-up	Show the equation, masks, CRS, resampling, robust-fit settings, residual plot and the exact train/test split.

Speaking rules

- Start with the problem and output, not team-member introductions or a list of technologies.
- Look at judges when explaining; look at the screen only while pointing to evidence.
- Do not pass the laptop between six people. One operator controls the demo while speakers hand off verbally.
- If you do not know, say: "We have not verified that yet. Our test would be..." Then describe the test.
- Use the same terms in speech, UI, deck and handout: rDSM, metric DSM, reference, units and validation.

16. One-page handout blueprint

The handout is not a miniature report. It should let a judge reconstruct the full answer at a glance.

Zone	Content
Top 20%	TerraFly name, PS ID/title, one-sentence claim and one before/after visual.
Left 25%	Problem, users, two modes and why existing sensing is not always available.
Center 35%	Five-step architecture: image - depth - calibration - DSM - textured 3D viewer.
Right 20%	Measured results table: RMSE/MAE/correlation, runtime, landscapes and build status. Leave blanks until measured.
Bottom strip	Novelty, limitations, offline/data statement, QR to demo/repository and team ownership.

THE HANDOUT TEST

Cover the title. A judge should still be able to answer: What goes in? What comes out? When are metres valid? How was it tested? What is different? What fails?

Do not include

- Dense technology logos without explaining their role.
- Future features presented as if implemented.
- An accuracy percentage without metric, units and dataset.
- A long paragraph about national security that makes unverified policy claims.
- QR codes that require a login or internet connection at the venue without a local fallback.

17. Final 48-hour evidence checklist

Area	Ready when...
Core path	One known PNG/JPG and one known GeoTIFF complete without manual code edits.
Truth labels	Relative mode never shows metres; metric mode records reference, CRS and units.
Metrics	RMSE, MAE and correlation are reproduced from a saved command/script and held-out data.
Landscapes	At least a small, honest table covers available categories and clearly marks missing categories.
Viewer	Texture alignment, orbit, first-person, height/slope and export are checked.
Reliability	Bad file, missing CRS, GPU failure, large image and interrupted job return useful errors.
Packaging	A second machine/profile can launch from the README or packaged launcher.
Demo backup	Recording, screenshots and exports exist on two laptops and one phone/USB.
Research	Every external claim has a source; every dataset/model has license/provenance.
Team	Every member has spoken three full rehearsals and can explain their artifact.
Numbers	Slide, handout, speech and app show identical measured values.
Limitations	The team can name the three biggest failure conditions and the next experiment.

FINAL STANDARD

Do not try to appear like you know everything. Appear like a team that knows exactly what it built, what it measured, what it does not know and how it will test the next risk.

18. Sources and evidence ledger

[P1] Supplied SIH 2026 Official PS.xlsx - row SIH26175. Local source provided by the user; closing date shown as 20 Sep 2026.

[P2] Supplied SIH 2026 SIGCE Score Sheet.pdf - local internal result. Used only for aggregate score/theme observations; no personal data reproduced.

W1: MathWorks Student Lounge - From Real Roads to Real Simulations: How Team TwinX Won SIH 2025.

<https://blogs.mathworks.com/student-lounge/2026/04/06/from-real-roads-to-real-simulations-how-team-twinx-won-smart-india-hackathon-2025/>

W2: India Today - How students built a voice-based bus booking system and won SIH 2025.

<https://www.indiatoday.in/education-today/featurephilia/story/how-student-built-voice-based-bus-booking-system-and-won-the-smart-india-hackathon-2025-2835019-2025-12-12>

W3: Public SIH 2025 winner repository - Ourobonics / Kritrim.

https://github.com/Baksheeshs/ourobonics-kritrim_SIH_2025_Winner

W4: Winner-authored Reddit experience - how the team says it was shortlisted and won SIH 2025. Treat as experience, not official criteria.

https://www.reddit.com/r/Indian_Academia/comments/1uz483r/how_we_got_shortlisted_for_the_smart_india/

W5: SIH 2025 finalist experience. Treat as finalist advice, not national winner evidence.

https://www.reddit.com/r/bmsit/comments/1vlgof8/this_is_how_we_made_it_to_the_sih_2025_grand/

O1: Press Information Bureau - 36-hour SIH 2025 software edition conclusion at Nagpur nodal centre.

<https://www.pib.gov.in/PressReleasePage.aspx?PRID=2202893&=2&=48>

O2: JIMS institute internal-hackathon report - PPT round and prototype evaluation dimensions.

https://www.jimsindia.org/JIMS_Latest_Events.aspx?id=1606

I1: Department of Science and Technology - Guidelines for acquiring and producing geospatial data and services.

<https://geospatial.dst.gov.in/Guidelines.aspx>

I2: Survey of India - New Guidelines on Geospatial Data 2021.

<https://surveyofindia.gov.in/pages/new-guidelines-on-geospatial-data-2021>

I3: Google Earth Education FAQ - 3D imagery acquisition and local flight-regulation dependence.

https://www.google.com/intl/en_us/earth/education/faq/

I4: Google Earth Studio best practices - imagery detail varies by location.

<https://earth.google.com/studio/docs/best-practices/>

Research note: source availability and web pages can change. Save local copies or screenshots of the sources used in your final deck, record access date, and do not rely on a single article for a technical or policy claim.